TDWCNet: Triple UNet With Dual-Window Convolution for Hyperspectral Anomaly Detection

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***Abstract*— In recent years, deep learning technology has emerged as the primary research focus in the field of hyperspectral anomaly detection (HAD) and has demonstrated satisfactory detection performance. Existing deep learning-based methods mainly utilize reconstruction errors as criteria for anomaly detection. However, they lack effective suppression of anomaly information in the background reconstruction process, and encounter challenges in addressing scenarios involving the coexistence of multiscale anomalies, which limits the performance of HAD. In order to reconstruct clean and reliable background images, this article proposes a Triple-UNet with dual-window convolution called TDWCNet for HAD. Specifically, we introduce a dual-window convolution that shields pixels within the inner window and only utilizes pixels between the inner and outer windows to reconstruct the central pixel. Based on the dual-window convolution, we construct the DWCBlock module, which serves as the core component for background reconstruction. To address the coexistence of multiscale anomalies, the Triple-UNet structure is designed, which organically combines three DWCBlock modules to gradually eliminate abnormal pixels that are inadvertently reconstructed due to inappropriate convolution kernels. Furthermore, adaptive mean-squared error (mse) and structural similarity index (SSIM) losses are employed to suppress anomaly reconstruction. Extensive experiments conducted on four publicly available datasets demonstrate that TDWCNet achieves satisfactory detection performance. The codes of this work will be available for the sake of reproducibility at: https://github.com/szc2277/TDWCNet-HAD.**

***Index Terms*— Background reconstruction, dual-window convolution, hyperspectral anomaly detection (HAD), Triple-UNet.**

I. Introduction

HYPERSPECTRAL remote sensing technology has provided a powerful tool for Earth science and resource management by offering wealth spectral information, providing a new perspective on surface changes. In recent years, hyperspectral image (HSI) has received widespread attention owing to its spectral and spatial properties [1]. Contrasted with the visible spectrum, HSI systems capture intricate spectral details, facilitating nuanced understanding and recognition of target characteristics in a distinct manner [2].

Hyperspectral anomaly detection (HAD) refers to the identification of pixels in an image with spectral characteristics significantly different from adjacent or overall background pixels. HAD finds extensive real-world applications, such as military reconnaissance [3], [4], camouflage identification [5], fine agriculture [6], and mineral identification [7]. Given the absence of prior knowledge regarding the spectral characteristics of the target in HAD, detecting such anomalies poses a considerable challenge [8].

Various HAD methods have been proposed in recent decades. The traditional HAD methods adopt a statistical framework to model the background distribution and identify pixels deviating from this model as anomalies. For example, the Reed–Xiaoli (RX) method [9] assumes that the background obeys the Gaussian distribution and detects anomalies by calculating the Mahalanobis distance between test pixels and the estimated background model. However, the multivariate Gaussian distribution is not sufficient to model the entire background statistics [10], which limits its effectiveness in complex scenarios. Representation-based HAD methods offer an alternative approach, aiming to represent all pixels using a certain dictionary in a reasonable model, with the residual indicating the abnormal level of pixels [11]. For example, Li and Du [12] proposed a collaborative representation-based detector (CRD), employing a sliding dual-window strategy to construct a background dictionary that collaboratively represents the central pixel. Anomalies are then discerned by calculating the reconstruction error for each pixel. However, these representation-based methods often introduce a large number of parameters, which may result in excessive computational costs. Additionally, they typically overlook the local correlation of spatial information.

In recent years, deep learning has achieved significant breakthroughs in speech and image recognition, demonstrating promising prospects and emerged as a major research focus in the field of HAD. Presently, existing supervised HAD methods predominantly rely on similarity metrics. Li et al. [13] proposed a pioneering method called convolutional neural network detector (CNND), which applies convolutional neural

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networks to HAD for the first time. This method leverages images with classification labels obtained from the same sensor to generate pixel pair differences, yielding a large number of samples for training the network. Through this training, the network learns to extract discriminative features, facilitating the distinction of similarities between pixel pairs. However, CNND entails substantial computational overhead, and the labeled data used in training may be difficult to apply to many real-world systems.

Given the limitations of supervised methods, unsupervised HAD methods have received great attention from researchers because of their characteristics of not requiring labeled data. Within the domain of unsupervised deep learning, typical networks include autoencoder (AE) networks and generative adversarial networks (GANs), and many unsupervised HAD methods leveraging these networks and their variations have been proposed. They acquire the spectral characteristics of the background in HSI so that the network can reconstruct background pixels but not anomalies. Subsequently, the reconstruction result is then residualized with the original data, and the reconstruction error serves as the measure of anomaly level. Bati et al. [14] introduced AE into the field of HAD for the first time, which is of groundbreaking significance. By learning the background representation, the background image is reconstructed and then compared with the original image. Notably, a significant reconstruction error indicates a high probability of an anomaly. Jiang et al. [15] proposed an unsupervised discriminative reconstruction constrained GAN (HADGAN). Its main goal is to obtain discriminative background reconstruction while suppressing the influence of abnormal objects. SSCRSAE [16] incorporates collaborative representation (CR) constraints into stacked AEs to extract complex nonlinear features, and CRD is subsequently employed to obtain detection results.

The above methods ignore spatial correlation, focusing solely on reconstruction tasks within the spectral domain, which is unfavorable for anomaly detection. To address this limitation, many approaches leveraging CNN and transformer architectures have been developed to extract spatial information in HSI for background reconstruction. Wang et al. [17] constructed a fully convolutional AE (Auto-AD) to reconstruct the background image, and designed an adaptive weighted loss function to further suppress reconstruction of abnormal pixels. Nevertheless, Auto-AD performs excessive downsampling operations during the encoder stage, leading to the loss of valuable details required for the background reconstruction process. Lian et al. [18] introduced a gated transformer network for HAD (GT-HAD), which leverages the spatial–spectral similarity to model the features of background and anomalies, thereby augmenting background features and weakening anomaly features. However, the feature redefinition process of GT-HAD is performed at the patch level instead of the pixel level, which makes the boundaries of the detected anomaly targets less accurate. More importantly, these methods ignore the spatial correlation of anomaly during the background reconstruction process and fail to adequately suppress abnormal information, leading to the reconstruction of certain abnormal pixels and consequently influencing the accuracy of detection results.

To overcome this challenge, recent methods have emerged aiming to mitigate spatial correlations of anomaly during the background reconstruction process, prevent contamination by abnormal information, and solely utilize background information for pixel reconstruction. Gao et al. [19] proposed BS^{3}LNet, integrating the unconsidered-spot network from the realm of self-supervision into HAD.[1] This network reconstructs each pixel solely relying on its neighboring pixels, thereby facilitating realistic background reconstruction. BS^{3}LNet demonstrates that unconsidered-spot network is effective for small-sized anomalies, but it faces limitations when dealing with large-sized anomalies. Wang et al. [20] (BockNet's author group) expanded the scope of unconsidered-spots and constructed a self-supervised unconsidered-block network with a guard window, enhancing the capability of the unconsidered-spot network to detect large-sized anomalies. Wang et al. [21] introduced a corner-visible convolution aimed at eliminating spatial correlations with potential anomalies for background generation. This effectively prevents contamination of the receptive field by abnormal pixels, thus inhibiting the reconstruction of abnormal pixels. Although these methods demonstrate satisfactory detection performance, they encounter challenges when dealing with scenarios involving the coexistence of multiscale anomalies. The primary reason lies in their utilization of the conventional encoder–decoder architecture, which overly relies on the range of the receptive field when addressing multiscale anomalies. If the range of receptive field is too small, adjacent abnormal pixels may not be excluded during the reconstruction of large-sized anomalies. Conversely, if the range of receptive field is too large, the spatial correlation between background pixels within the receptive field will be weakened, hindering background reconstruction. By iteratively utilizing the encoder–decoder architecture for coarse-to-fine background reconstruction, the gradual elimination of potentially reconstructed abnormal pixels may alleviate the overreliance on receptive fields to some extent.

Therefore, the primary challenges of HAD are how to effectively suppress the reconstruction of abnormal pixels during the background reconstruction process to obtain a clean background image, and how to address scenarios involving the coexistence of multiscale anomalies. To address these challenges, this article proposes a Triple-UNet based on dual-window convolution (TDWCNet) to reconstruct the background in an end-to-end manner, where the anomalies are manifested as prominent reconstruction errors. This method limits the receptive field to the area between the inner and outer windows through dual-window convolution, which can effectively eliminate spatial correlations of anomaly and reduce the contamination of the receptive field by abnormal pixels, thus weakening the ability to reconstruct abnormal features. Fig. 1 illustrates the difference in receptive fields and reconstruction results between conventional convolution, unconsidered-spot network, and our dual-window convolution. The spatial correlation of the background is learned by mapping the uniform

[1]In this article, we use the phrase "unconsidered-spot" to be in line with IEEE non-inclusive language guidelines. In [19], the original phrase used is "blind-spot."

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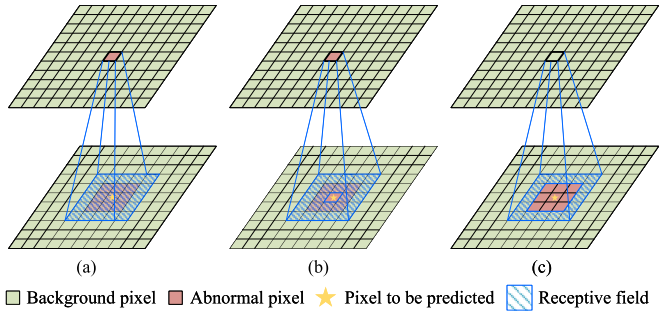


Fig. 1. Comparison of receptive fields and reconstruction results between conventional convolution, unconsidered-spot network convolution and dual-window convolution. (a) In conventional convolution, all pixels within the receptive field are utilized for prediction. (b) In unconsidered-spot network convolution, the receptive field excludes the center pixel (i.e., the pixel to be predicted), thereby preventing it from learning its own identity. (c) In dual-window convolution, the receptive field is the area between the inner and outer windows. When the pixel to be predicted is an abnormal pixel, the spatially related abnormal information in the central area is effectively eliminated.

noise space to the hyperspectral space. In order to cope with the coexistence of multiscale anomalies, the Triple-UNet structure is devised, facilitating the adaptation of the same convolution kernel to anomaly targets with varying sizes. Through this network, abnormal pixels that are inadvertently reconstructed can eventually be restored as background pixels. In addition, a weighted adaptive loss function is employed to further suppress anomaly reconstructions.

The main contributions of this article can be summarized as follows.

1) A dual-window convolution is proposed to limit the receptive field to the area between the inner and outer windows, and only use background pixels far away from the central area for reconstruction. By adjusting the size of the inner and outer windows, the inner window serves as a shielding area, effectively reducing the pollution caused by abnormal pixels in the central area.
2) A reconstruction-based Triple-UNet network (TDWC-Net) is proposed for unsupervised HAD. The network utilizes dual-window convolution to learn background features to reconstruct the background image. At the same time, it can effectively eliminate anomalies that are inadvertently reconstructed due to abnormal pixels contaminating the receptive field, thereby coping with the coexistence of multiple scale anomalies.
3) Extensive experiments were conducted on four HSI datasets with anomalies of different sizes to evaluate the performance of our method. The experimental results show that our method has accurate detection results for abnormal targets of different sizes in different scenarios, and achieves state-of-the-art accuracy for unsupervised HAD.

The remainder of the article is organized as follows. Section II describes some related works. Section III introduces the proposed HSI anomaly detection method in detail. Section IV presents our extensive experimental results and compares our method with other existing methods. Finally, Section V concludes the article.

II. Related Work

At present, existing HAD methods can be roughly divided into three categories, namely statistical-based methods, representation-based methods, and deep learning-based methods, and these methods have shown satisfactory performance.

A. Statistical-Based Methods

In the field of HAD, the methods based on statistical theory are the earliest and most classic methods. The RX algorithm [9] was proposed in 1990 and is considered the benchmark for this type of method. It assumes that the background follows a Gaussian distribution and estimates the probability density function by calculating the covariance and mean vectors of all background samples. Anomalies are detected by calculating the Mahalanobis distance between the test pixel and the background model. The detector can be expressed as

$$D_{\mathrm{RX}}(\mathbf{x}) = (\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}) \tag{1}$$

where $\mathbf{x}$ is the spectral vector of the testing pixel, $\boldsymbol{\mu}$ is the estimated background mean, and Σ is the estimated background covariance matrix. However, the detection performance of the RX method is limited because the Gaussian distribution is not sufficient to model the entire background statistics. Based on the RX method, many improved methods have been proposed, such as the local RX method [22], the kernel RX method [23], the cluster kernel RX method [24], and the weighted RX method [25]. But these methods still rely on specific model assumptions about the statistical distribution of the background. However, the complexity of hyperspectral data and the diversity of ground cover types make it difficult to satisfy the assumption of a single statistical distribution model, which may lead to serious false alarms.

B. Representation-Based Methods

The core idea of the representation-based methods is to represent each pixel as a linear combination of other pixels by learning the representation mode of the data, and the residual is the abnormality degree of the pixel. Such representation models usually assume that normal pixels have a sparse or low-rank property, while abnormal pixels may have more complex representation patterns. The representation model can be expressed as

$$\mathbf{x} \approx \mathbf{D}\boldsymbol{\alpha} \tag{2}$$

where $\mathbf{x}$ is a pixel signal, $\mathbf{D}$ is a dictionary, and $\boldsymbol{\alpha}$ is a representation coefficient vector. Depending on the regularization constraints on $\boldsymbol{\alpha}$, HAD based on reconstruction is primarily categorized into sparse representation (SR) [26], [27], CR [28], [29], and low-rank representation (LRR) [30], [31]. Both SR and CR anomaly location methods are pixel-by-pixel detection, while LRR describes the global structure of data [32], taking into account the low-rank properties of backgrounds and the sparse properties of anomalies [33], [34]. Some representative methods include the low-rank and SR detector (LRASR) [35], low-rank matrix decomposition based

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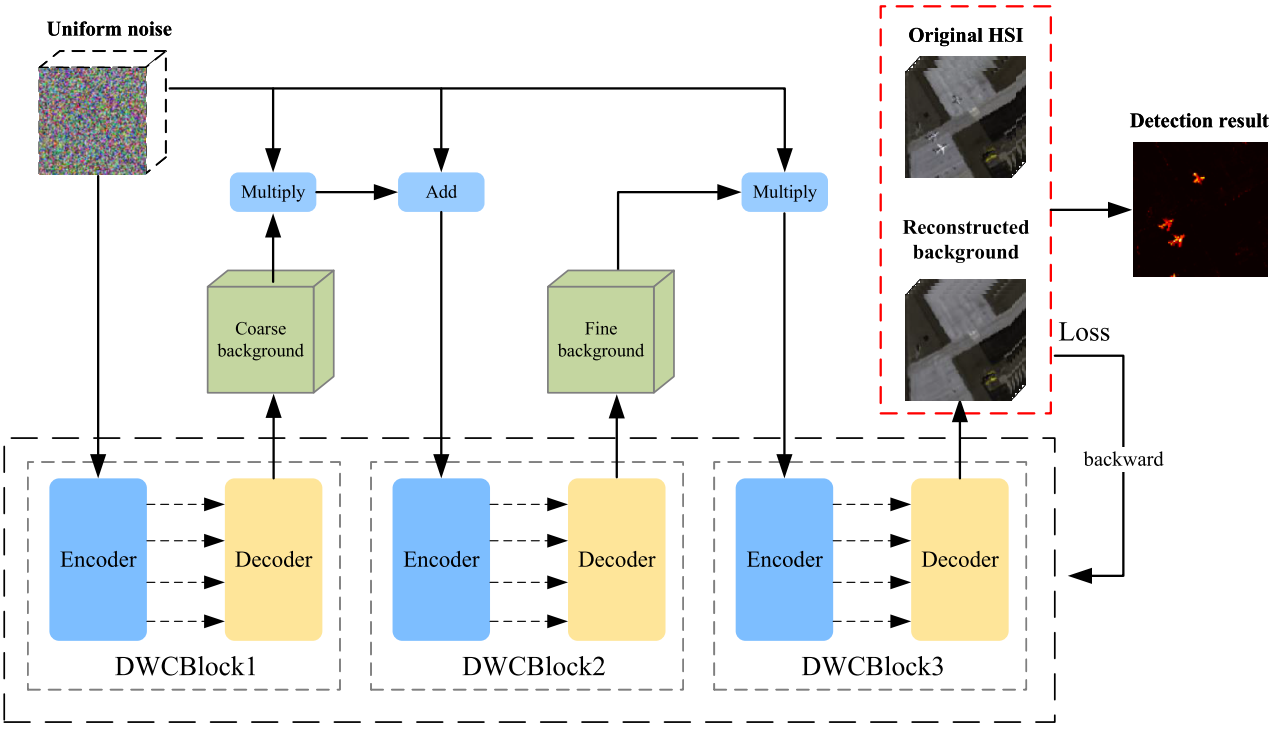


Fig. 2. Overall structure of the proposed TDWCNet for HAD.

on Mahalanobis distance for anomaly detection (LSMAD) [36], low-rank and CR detector (LRCRD) [37], and so on. Such methods have shown strong detection capabilities, but they suffer from the complexities of feature engineering, limited generalization across diverse datasets, and computational intensity, impacting their efficiency and applicability in real-world scenarios.

C. Deep Learning-Based Methods

Deep learning technology can extract deep features and obtain more data features than traditional methods, so it has received widespread attention. Existing deep learning-based HAD methods can be subdivided into similarity measure-based methods and reconstruction-based methods.

Methods based on similarity measures [13], [38] usually select a large number of pixel pairs in the classified HSI with labels to form samples. If two pixels come from the same class label, the sample they constitute is labeled 0, otherwise it is labeled 1. These pixel pairs are subsequently utilized to train the network, enabling the network to distinguish similarities between pixel pairs. The degree of abnormality of the current pixel is expressed by calculating the difference score between each pixel to be measured and its surrounding pixels. However, the detection performance of this type of method is affected by the quality of the constructed samples and requires a lot of time.

Reconstruction-based methods usually use AE networks, GANs, and their variants to construct reconstruction networks, which reconstruct the background components by learning the spectral features of the background in the original HSI [14], [15], [17], [39], [40], [41]. Therefore, anomalies often appear as large reconstruction errors. Finally, Mahalanobis distance or other post-processing is often applied to obtain detection results. Recently, the transformer-based background reconstruction methods have shown promising results. Applying transformers to HAD for the first time, Xiao et al. [42] proposed a HAD method with a spatial–spectral dual-window mask transformer, which extract features from global and local perspectives, and suppress the reconstruction of anomaly targets adaptively. Wu and Wang [43] proposed a method based on 3-D-transformer, which utilizes transformer modules in both spatial and spectral dimensions sequentially to reconstruct each pixel. This type of method does not require labeled data, is more consistent with HAD tasks, and achieves satisfactory results. However, they often fall short in fully exploring spatial information and lack effective suppression of abnormal information during the reconstruction process.

III. Proposed Method

Based on the exploration of the spatial and spectral characteristics of background and abnormal targets in HSI, we designed TDWCNet to detect abnormal targets of various sizes across different scenarios. Our model is trained in an unsupervised manner without distinguishing between training and test sets. It aims to reconstruct the background image of the input HSI, using the reconstruction error between the background and the original image to identify pixel anomalies. As depicted in Fig. 2, the TDWCNet is composed of three DWCBlock modules with identical structures. Each DWCBlock consists of an encoder and a decoder interconnected by skip connections. The input of the network is the uniform noise space $\mathbf{X}_o \in \mathbb{R}^{H\times W\times C}$, which is mapped to the HSI background space $\mathbf{X}_{\text{bg}} \in \mathbb{R}^{H\times W\times C}$ through the fully convolutional network, where H and W denote the height and width of the image, respectively, and C represents the number

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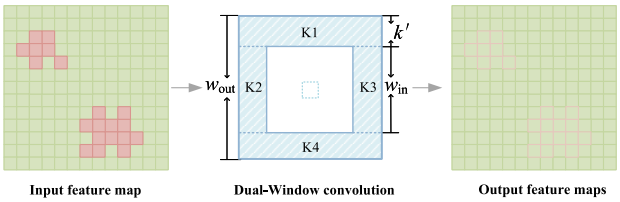


Fig. 3. Illustration of dual-window convolution and its convolution result, where the red grid represents anomalies in the input, the two blue windows represent the inner and outer windows of the dual-window convolution, and the shadow between windows represents the receptive field of the convolution.

of spectral bands. To obtain a pure background image, the network utilizes dual-window convolution to extract background features and mitigate anomaly contamination. Simultaneously, we introduced a Triple-UNet structure to eliminate abnormal pixels that are accidentally reconstructed, enabling the same convolution kernel to adapt to abnormal targets of varying sizes. Finally, the reconstruction error indicates the anomaly level of each pixel.

A. Dual-Window Convolution

To achieve pixel-level background reconstruction on the original HSI, for each pixel to be reconstructed x_i, our goal is to effectively utilize the surrounding background pixels, learn the feature representation of the background, and reconstruct it as a background pixel $\hat{x}_i$. Considering the spatial correlation of anomalies, and aiming to effectively reduce the contamination of the receptive field by abnormal pixels, we propose dual-window convolution to learn the representation of background features. Fig. 3 illustrates the structure of dual-window convolution and the result of using it to convolve a feature map containing anomalies.

As shown in Fig. 3, the receptive field of dual-window convolution is the region between the inner and outer windows, which can be changed by adjusting the size of the two windows. The inner window serves as a shielded region, preventing the involvement of pixels within it in the convolution operation. In cases where the central pixel is an abnormal pixel, its neighborhood tends to contain more related abnormal pixels due to the spatial correlation. By appropriately adjusting the size of the inner window, both the central pixel and its adjacent abnormal pixels can be effectively shielded, thus minimizing the contamination of the receptive field by abnormal pixels. Specifically, the dual window refers to an inner window with a size of $w_{\text{in}} \times w_{\text{in}}$ and an outer window with size of $w_{\text{out}} \times w_{\text{out}}$ (both w_{in} and w_{out} are odd numbers), where the inner window is centered on the outer window. The receptive field of dual-window convolution is divided into four disjoint regions called sub-kernels $\mathbf{k}_l, \forall l \in \{1, 2, 3, 4\}$. Among them, the shape of $\mathbf{k}_1$ and $\mathbf{k}_4$ is $k' \times w_{\text{out}} \times c$, and the shape of $\mathbf{k}_2$ and $\mathbf{k}_3$ is $w_{\text{in}} \times k' \times c$, where $k' = (w_{\text{out}} - w_{\text{in}})/2$ and c is the number of input channels. For any pixel $x_{i,j}$, take it as the center of the dual-window, and then perform dual-window convolution on it. The obtained convolution result is expressed as

$$\hat{x}_{i,j} = \sum_{l=1}^{4} \left(\mathbf{K}_l * \mathbf{X}_{i,j,l} \right) \tag{3}$$

where $\mathbf{K}_l$ represents the convolution kernel with the same shape as region $\mathbf{k}_l$, $\mathbf{X}_{i,j,l}$ denotes the lth region within the receptive field centered of $x_{i,j}$, and $*$ is matrix multiplication.

Compared with traditional convolution, dual-window convolution considers the spatial correlation of anomalies and effectively blocks the pixel to be reconstructed itself and its nearest neighboring pixels by shielding the pixels within the inner window. When the central pixel is abnormal, dual-window convolution effectively reduces the contamination of the receptive field by abnormal pixels, and utilizes background pixels far away from the central region to learn the feature representation of the background, which is essential for reconstructing a pure background.

B. Reconstruction Module Based on Dual-Window Convolution (DWCBlock)

Building upon dual-window convolution, we construct a simplified U-Net module termed DWCBlock for background reconstruction. The detailed network structure is illustrated in Fig. 4. To avoid the reduction of spatial resolution caused by downsampling operations, which might result in the loss of crucial details required for background reconstruction, we maintain the size of the feature map during encoding to be consistent with the input.

The encoder consists of stacked N encoder blocks; each block is composed of a dual-window convolution layer, a batch normalization (BN) layer, and a rectified linear unit (ReLU) activation function. The designed double-window convolution shields the pixels in the inner window, utilizing only those distant from the central area to capture background features. This effectively prevents the network from learning potential anomalies and weakens the network's ability to express abnormalities. Employing BN and ReLU activation function following dual-window convolution not only expedites the training process and enhances the model's generalization capabilities, but also boosts the network's nonlinear characteristics and elevates its expressive power. The input to the network is defined as $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, and the output of each encoder block is defined as $\mathbf{F}_{\text{e}}^{i} \in \mathbb{R}^{H \times W \times c}$, where $i \in \{1, 2, \ldots, N\}$ is the number of encoder blocks and c is the number of channels.

To eliminate the continuity loss of information caused by repeated stacked dual-window convolutions, the decoder block employs CBAM to help the model focus on important feature regions and utilizes standard convolution layers to capture local detail information. The input of the first decoder block $\mathbf{F}_{\text{d}}^{1} \in \mathbb{R}^{H \times W \times c}$ is obtained from $\mathbf{F}_{\text{e}}^{N}$ and skip connection

$$\mathbf{F}_{\text{d}}^{1} = \text{concat}\left(\mathbf{F}_{\text{e}}^{N}, \mathbf{Skip}\left(\mathbf{F}_{\text{e}}^{N-1} \right) \right) \tag{4}$$

$$\mathbf{Skip}\left(\mathbf{F}_{\text{e}}^{N-1} \right) = \text{ReLU}\left(\text{BN}\left(\text{conv}_{1\times1}\left(\mathbf{F}_{\text{e}}^{N-1} \right) \right) \right). \tag{5}$$

The skip connection serves to retain detailed information from the network's initial stages and fuse this information with higher-level features generated by the decoder blocks, thereby enhancing the model's ability to simultaneously capture fine-grained details and high-level context. Each decoder block initially employs a 1×1 convolution to halve the number of channels, and then passes through CBAM. CBAM

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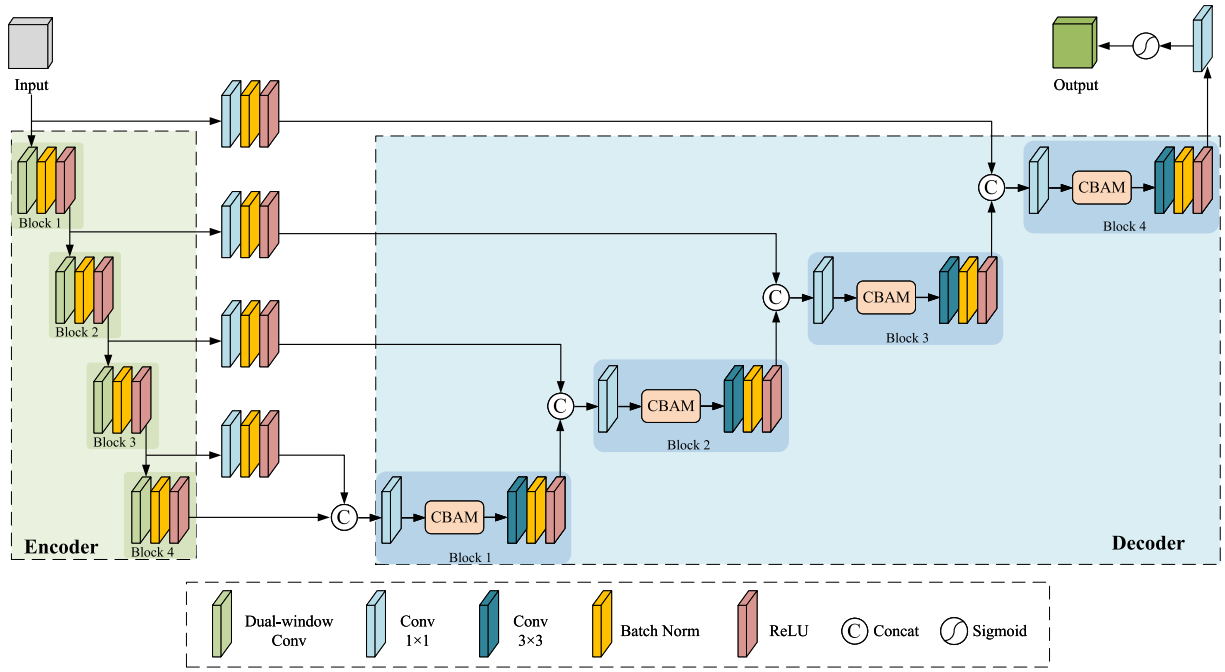


Fig. 4. Detailed network structure of DWCBlock.

explicitly models the inter-channel and spatial interdependencies, allowing the network to allocate more attention to important features and regions during the decoding process, thereby enhancing the network's capacity to represent background features. Subsequently, a 3×3 convolution with BN layer and ReLU activation function is employed to further extract and integrate the fused features. The input of the ith decoder block can be expressed as

$$\mathbf{F}_{\mathrm{d}}^{i} = \mathrm{concat}\left(\mathbf{De}\left(\mathbf{F}_{\mathrm{d}}^{i-1}\right), \mathbf{Skip}\left(\mathbf{F}_{\mathrm{e}}^{N-i}\right)\right) \tag{6}$$

$$\mathbf{De}\left(\mathbf{F}_{\mathrm{d}}^{i-1}\right) = \mathrm{conv}_{3\times3}\left(\mathrm{CBAM}\left(\mathrm{conv}_{1\times1}\left(\mathbf{F}_{\mathrm{d}}^{i-1}\right)\right)\right) \tag{7}$$

where $i \in \{2, \ldots, N\}$. When i equals N, $\mathbf{F}_{\mathrm{e}}^{0}$ is the original input to the network.

Following the final decoder block, a 1×1 convolution is utilized to adjust the number of channels, and then a sigmoid function is employed to limit the value of the output image between 0 and 1. Consequently, the output of the network can be expressed as

$$\hat{\mathbf{X}} = \mathrm{Sigmoid}\left(\mathrm{conv}_{1\times1}\left(\mathbf{De}\left(\mathbf{F}_{\mathrm{d}}^{N}\right)\right)\right) \tag{8}$$

C. Triple-UNet Structure

Abnormal targets in HSI typically exhibit unpredictable distributions and often vary significantly in size. As illustrated in Fig. 5(a), when the same convolution kernel is applied to abnormal targets of different sizes, due to the spatial correlation of anomalies, the receptive field will inevitably be contaminated by anomalous pixels when reconstructing the background of anomalous pixels in larger-size abnormal target. Fig. 5(b) shows the results of reconstructing them using DWCBlock.

To address this issue, a straightforward method is to enlarge the size of the inner window so that it can completely shield the abnormal pixels. However, while effectively excluding anomalies, an excessively large inner window may inadvertently result in the loss of crucial background information. Furthermore, enlarging the inner window expands the scope of receptive field, which weakens the spatial correlation between background pixels within the receptive field, impeding background reconstruction.

To tackle this challenge, we designed the Triple-UNet structure, which organically combines three DWCBlock modules to form the final network TDWCNet, as depicted in Fig 2. Specifically, the TDWCNet consists of three DWCBlock modules with identical structure, each of which can be independently used as a background reconstruction network. The input of the first DWCBlock module is uniform noise $\mathbf{X}_o \in \mathbb{R}^{H\times W\times C}$, and the reconstruction result is a rough background image, because, as shown in Fig. 5(b), some abnormal pixels in large-size abnormal targets may be reconstructed. Next, the reconstruction result undergoes pointwise multiplication with the initial input, and the result is added to the initial data to yield the input of the second DWCBlock module. By multiplying the input noise, the initial background features are integrated with the noise, ensuring that any strong anomalous signals are present but weakened. The subsequent addition of noise preserves the structure and variability of the original noise, ensuring that the second DWCBlock receives a richer input that enhances the background information while suppressing the anomalous information.

Subsequently, the second DWCBlock module is employed for a secondary background reconstruction to achieve a more refined background image. However, as shown in Fig. 5(c), there may still be reconstructed abnormal pixels from large-size abnormal targets in the reconstruction result. Therefore, we utilize the reconstruction result to multiply pointwise with the initial input again to obtain the input of the third

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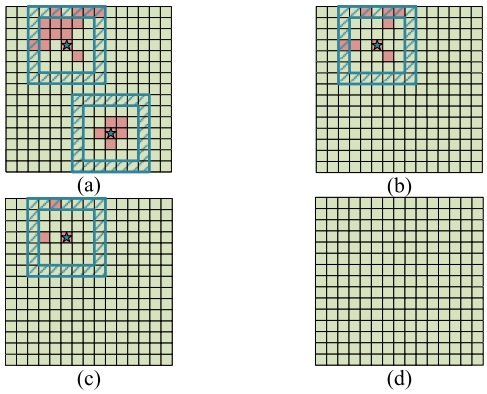


Fig. 5. Illustrations of the background images obtained after the input feature map containing abnormal targets of different sizes passes through the three DWCBlock modules in the TDWCNet in sequence. The red grid represents abnormal pixels, the blue star represents the current pixel to be reconstructed, and the blue shaded area represents the receptive field of the dual-window convolution. (a) Input feature maps with abnormal targets of different sizes. (b) Reconstruction result of the first DWCBlock module (rough background). (c) Reconstruction result of the second DWCBlock module (fine background). (d) Final reconstruction result.

DWCBlock module. This operation further suppresses the abnormal information while enhancing the background structure. The background image is further refined by maintaining the background information without reintroducing the noise component, ensuring that the input to the third DWCBlock is a more refined background.

Following three background reconstructions, a pure background image $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times C}$ is obtained. Fig. 5 displays the results of three background reconstructions using TDWCNet on input containing abnormal objects of different sizes.

Finally, we used ℓ_2-norm to calculate the anomaly score

$$\mathbf{D} = \|\mathbf{X} - \hat{\mathbf{X}}\|_2. \tag{9}$$

D. Loss Function

The goal of our method is to ensure that the background pixels in the generated image are similar to those in the original HSI through iterative training, while suppressing the similarity of abnormal pixels. Our method follows the loss function settings of CVNet [21], using adaptive mean-squared error (mse) loss and adaptive structural similarity index (SSIM) loss to optimize the network parameters.

The reconstruction error of each pixel can be calculated using the following:

$$e_{i,j} = \|\mathbf{x}_{i,j} - \hat{x}_{i,j}\|_2 \tag{10}$$

where $\mathbf{x}_{i,j} \in \mathbb{R}^C$ represents the pixel vector at position (i, j) in the original HSI, $\hat{x}_{i,j} \in \mathbb{R}^C$ represents the corresponding reconstructed output vector. The map of reconstruction errors E can be expressed as

$$\mathbf{E} = \left[e_{1,1}, \ldots, e_{1,W}; \ldots; e_{H,1}, \ldots, e_{H,W}\right]. \tag{11}$$

The weighted map W is then computed as follows:

$$w_{i,j} = \max(\mathbf{E}) - e_{i,j} \tag{12}$$

$$\mathbf{W} = \left[w_{1,1}, \ldots, w_{1,W}; \ldots; w_{H,1}, \ldots, w_{H,W}\right]. \tag{13}$$

The final loss function is defined as

$$\mathcal{L} = \text{mse}\left(\mathbf{WX}, \mathbf{W}\hat{\mathbf{X}}\right) + \lambda\left(1 - \text{SSIM}\left(\mathbf{WX}, \mathbf{W}\hat{\mathbf{X}}\right)\right) \tag{14}$$

where the parameter λ is used to control the balance between the two loss functions.

Algorithm 1 TDWCNet

Input: hyperspectral image $\mathbf{X} \in \mathbb{R}^{\mathbf{H}\times\mathbf{W}\times\mathbf{C}}$, parameters: 1) outer window size w_{out} and inner window size w_{in} and 2) loss function term λ

Output: detection result $\mathbf{D} \in \mathbb{R}^{H\times W}$

```
Initiate the input noise X_o ∈ R^{H×W×C} and weight map W ∈ R^{H×W};
while iterations < 1000 do
    Network forward;
    if iterations %200 = 0 then
        Update the weight map using Eqs.10-13;
    end
    Calculate loss using Eqs.14;
    Network backward;
end
Obtain the reconstructed background image X̂ ∈ R^{H×W×C};
Calculate the detection result using 9.
```

IV. Experimental Results and Analysis

In this section, we conduct comprehensive experiments on four real HSI datasets to verify the validity of the proposed TDWCNet. Additionally, by comparing our method with 10 existing state-of-the-art HAD technologies, we demonstrated that TDWCNet outperforms existing methods.

A. Datasets Description

1) Gulfport Dataset: It was acquired utilizing the AVIRIS sensor from Gulfport, which features a spatial resolution of 3.4 m/pixel. The image has a spatial size of 100 × 100 and encompasses 191 spectral bands. The anomalies come from three airplanes in various sizes with 60 abnormal pixels, which occupy 0.60% of the total number of pixels.

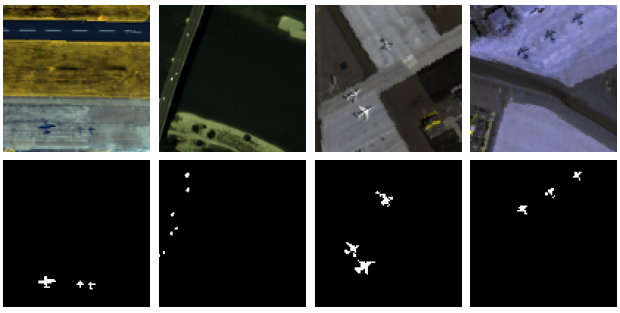
2) Pavia Dataset: It was acquired using the reflective optics system imaging spectrometer (ROSIS) sensor. The image has a spatial size of 150 × 150 and exhibits a spatial resolution of 1.3 m/pixel, along with 102 spectral bands. The primary anomalies are several vehicles on the bridge, encompassing 68 abnormal pixels, which correspond to 0.30% of the total number of pixels.

3) San Diego I Dataset: It was captured using the airborne visible/infrared imaging spectrometer (AVIRIS) at San Diego Airport, San Diego, CA, USA. The image has a spatial size of 100 × 100 and exhibits a spatial resolution of 3.5 m/pixel, along with 189 spectral bands. The anomalies came from three aircrafts in different directions, and the number of anomalous pixels was 134, accounting for 1.34% of the total number of pixels.

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TABLE I
OPTIMAL PARAMETERS OF ALL HAD METHODS ON FOUR DATASETS

Methods	Parameters	Gulfport	Pavia	San Diego I	San Diego II
RX	/	/	/	/	/
KRX	γ	1.5	3	2.5	1.5
CKRX	(n, γ)	$(5, 10^{-2})$	$(15, 10^{-5})$	$(5, 10^{-1})$	$(10, 10^{-6})$
CRD	$(\lambda, w_{out}, w_{in})$	$(10^{-6}, 19, 17)$	$(10^{-6}, 7, 5)$	$(10^{-6}, 17, 15)$	$(10^{-6}, 17, 15)$
GTVLRR	(λ, β, γ)	(0.5, 0.2, 0.05)	(0.5, 0.2, 0.05)	(0.5, 0.2, 0.05)	(0.05, 0.2, 0.05)
Auto-AD	σ	1.2×10^{-5}	1×10^{-5}	1.4×10^{-5}	1.4×10^{-5}
PDBSNet	$(l_r, f_{train}, f_{test})$	$(10^{-4}, 5, 5)$	$(10^{-4}, 3, 3)$	$(10^{-4}, 5, 5)$	$(10^{-4}, 4, 4)$
DCAE	(n, λ)	$(4, 0.1)$	$(3, 0)$	$(5, 0)$	$(4, 0.1)$
GT-HAD	/	/	/	/	/
UADNet	$(K, \lambda_{\mathrm{MSE}}, \lambda_{\mathrm{R}}, \lambda_{\mathrm{MTV}}, \lambda_{\mathrm{E}}, \lambda_{\mathrm{C}})$	$(3, 10^{-2}, 0.2, 0.1, 10^{-4}, 10^{-4})$	$(1, 0.5, 0.8, 0.1, 10^{-2}, 10^{-4})$	$(6, 0.5, 0.2, 0.1, 10^{-3}, 10^{-4})$	$(5, 0.5, 0.3, 0.1, 10^{-3}, 10^{-3})$
TDWCNet	$(w_{out}, w_{in}, \lambda)$	$(5, 3, 1)$	$(7, 3, 0.005)$	$(7, 5, 0.5)$	$(7, 3, 0.1)$



(a) (b) (c) (d)

Fig. 6. Pseudo-color images and ground truth maps of the four considered HSI datasets. (a) Gulfport. (b) Pavia. (c) San Diego I. (d) San Diego II.

4) San Diego II Dataset: San Diego I and II datasets are derived from different regions of the same original HSI, with the same spatial size, spatial resolution, and spectral band number. The anomaly came from three planes in the upper right corner, and the number of anomalous pixels was 58, accounting for 0.58% of the total number of pixels.

The above four real HSI datasets contain anomalous targets of different sizes, and their pseudo-color images and the ground truth maps of the targets are shown in Fig. 6.

B. Detection Results

C. Experiment Setup

1) Comparison Methods: To evaluate the performance of the proposed TDWCNet, we compare it with 10 existing HAD methods, including RX [9], KRX [23], CKRX [24], CRD [12], GTVLRR [44], Auto-AD [17], PDBSNet [45], DCAE [46], GT-HAD [18], and UADNet [47]. Among them, RX, KRX, and CKRX are statistics-based method; CRD and GTVLRR are representation-based methods; and Auto-AD, PDBSNet, DCAE, GT-HAD, and UADNet are deep learning-based methods. Specifically, Auto-AD, DCAE, and GT-HAD are unsupervised background reconstruction networks that utilize the residual difference between the original HSI and the reconstructed HSI to measure pixel abnormality, following the same principle as our method. PDBSNet trains the network to reconstruct the background in a self-supervised manner, while UADNet is an end-to-end algorithm that jointly performs hyperspectral unmixing and HAD based on deep clustering.

2) Parameter Settings: In the experiment, each DWCBlock module integrated four repeated encoder blocks and decoder blocks. The training of the network is optimized using the Adam optimizer with a learning rate of 0.01. The spectral band reduction ratio is set to 0.6. The iterative learning process is performed for 1000 epochs, and the weights in the loss function are adaptively initialized every 200 epochs.

For the 10 comparison methods, we adjust the parameters of all comparison methods through experiments and select the optimal parameters for comparative experiments. Table I summarizes the optimal parameters of 10 HAD methods and the proposed TDWCNet on 4 datasets.

3) Evaluation Metrics: To evaluate the detection performance of different methods, we considered the following four metrics: receiver operating characteristic (ROC) curve [48], area under the ROC curve (AUC) [49], the precision–recall curve (PRC) [50], [51], and Box–Whisker plot [52]. The ROC curve is plotted by true positive rate (TPR) versus false positive rate (FPR) at various thresholds (τ), which can qualitatively determine the superiority among different methods under the same false alarm condition. The AUC value obtained by calculating the whole area under the ROC curve can quantitatively illustrate the detection performance. The closer the AUC value of (TPR, FPR) is to 1, the better the detection accuracy. Since the numbers of positive and negative samples are imbalanced in the HAD task, HAD can be considered an imbalanced problem, and the PRC can characterize the performance of different algorithms under imbalanced scenarios. The PRC is plotted by precision versus recall at various thresholds (τ), allowing for the qualitative determination of the superiority among different methods under the same recall condition. Box–Whisker can visually reflect the separability between anomalies and background. The red box indicates the distribution of anomalous pixels, while the blue box indicates the distribution of background pixels. The distance between the red and blue boxes indicates the separability of background and anomalies. The height of the background box reflects the degree of background suppression, with smaller heights indicating better suppression effects on the background.

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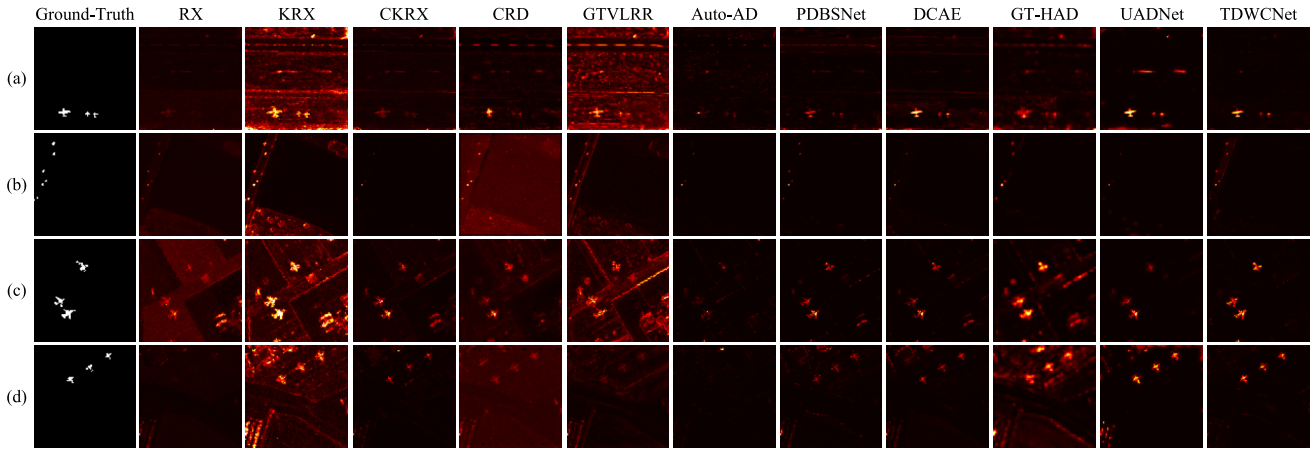


Fig. 7. Color anomaly maps of different HAD methods on four datasets. (a) Gulfport. (b) Pavia. (c) San Diego I. (d) San Diego II.

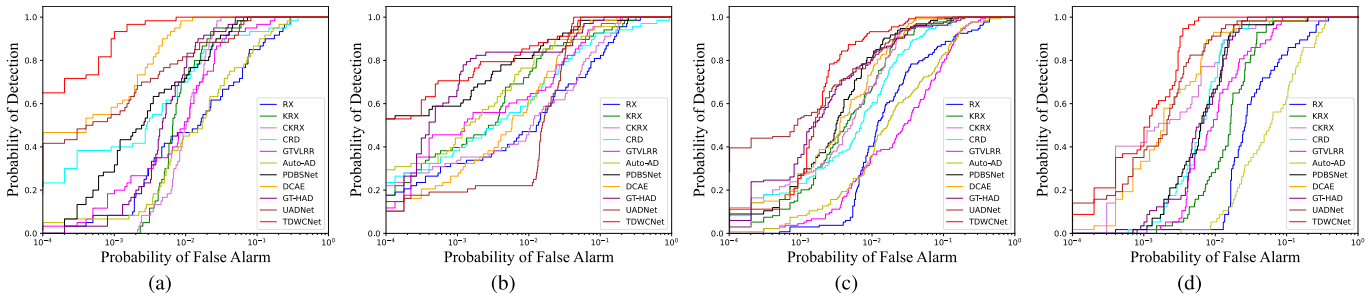


Fig. 8. ROC curves of different HAD methods on four datasets. (a) Gulfport. (b) Pavia. (c) San Diego I. (d) San Diego II.

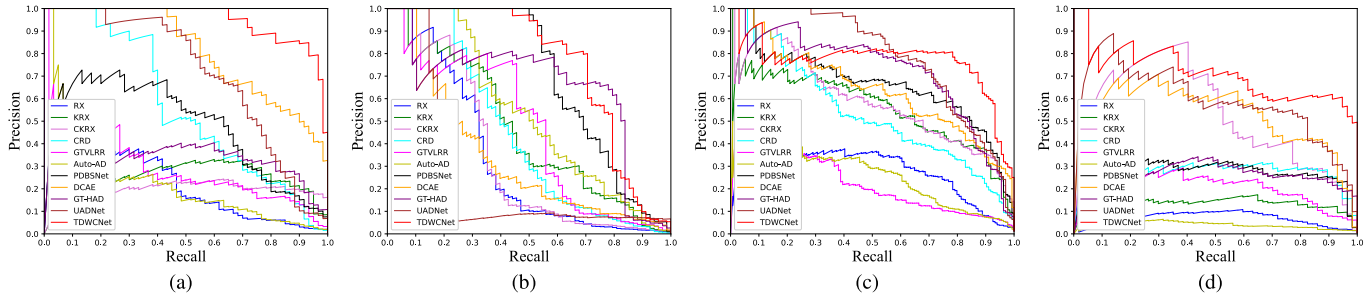


Fig. 9. PR curves of different HAD methods on four datasets. (a) Gulfport. (b) Pavia. (c) San Diego I. (d) San Diego II.

4) Gulfport: The abnormal targets in this dataset are three aircrafts with significant differences in size, and the integrity and prominence of the targets are both crucial, which poses a challenge to HAD methods. The detection maps of all HAD methods on the Gulfport dataset are shown in Fig. 7(a). It can be observed that the proposed TDWCNet shows strong performance in background suppression and highlights the integrity and salience of abnormal targets. Furthermore, the detection result is closely consistent with the ground truth, indicating that anomalies can be reliably distinguished from the background.

Comparatively, other methods cannot achieve such performance. The detection results of RX, CKRX, and Auto-AD are not satisfactory as they could barely detect the location of abnormal aircrafts. Although KRX and GTVLRR can highlight abnormal targets well, they still retain a significant amount of background information, such as dotted lines on the road, which leads to obvious false detections. Although CRD, DCAE, and UADNet can highlight abnormal targets, they still delineate some background targets. PDBSNet and GT-HAD are not sufficiently effective in highlighting abnormal targets. Additionally, GT-HAD's detected abnormal targets have overly large outlines with unclear boundaries.

The ROC curve in Fig. 8(a) and the PR curve in Fig. 9(a) clearly show that the curves of TDWCNet are always higher than those of other methods, indicating that TDWCNet's detection performance is significantly superior to other methods. In addition, TDWCNet achieves the highest $\mathrm{AUC}_{(P_d,P_f)}$ value in Table II. Furthermore, as shown in the Box–Whisker plot in Fig. 10(a), the anomaly and background boxes of KRX and GTVLRR are both relatively high, indicating that while effective at highlighting anomalies, their capability to suppress background is limited, which is consistent with the visual perception from the color anomaly maps. On the other

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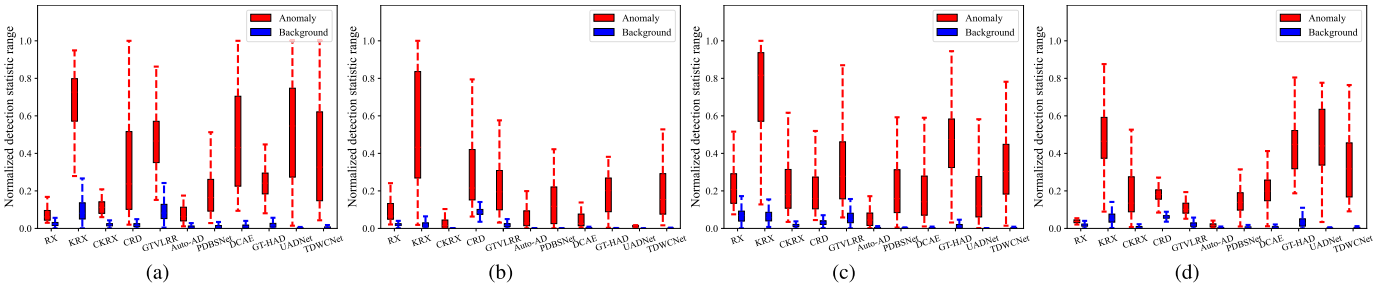


Fig. 10. Box–Whisker plots of different HAD methods on four datasets. (a) Gulfport. (b) Pavia. (c) San Diego I. (d) San Diego II.

TABLE II
$\mathrm{AUC}_{(P_d,P_f)}$ ↑ OF DIFFERENT HAD METHODS ON FOUR DATASETS

Datasets	RX	KRX	CKRX	CRD	GTVLRR	Auto-AD	PDBSNet	DCAE	GT-HAD	UADNet	TDWCNet
Gulfport	0.9526	0.9890	0.9880	0.9751	0.9814	0.9582	0.9905	0.9982	0.9915	0.9914	**0.9996**
Pavia	0.9538	0.9787	0.9628	0.9527	0.9802	0.9837	0.9908	0.9849	0.9907	0.9820	**0.9944**
San Diego I	0.9513	0.9885	0.9901	0.9827	0.9429	0.9476	0.9921	0.9919	0.9899	0.9908	**0.9964**
San Diego II	0.9398	0.9781	0.9882	0.9914	0.9824	0.9036	0.9883	0.9932	0.9915	0.9960	**0.9984**
Average	0.9494	0.9836	0.9823	0.9755	0.9717	0.9483	0.9904	0.9921	0.9909	0.9901	**0.9972**

hand, except for CKRX, DCAE, GT-HAD, and TDWCNet, other methods exhibit overlap between their background and anomaly boxes, indicating unsatisfactory background-anomaly separation capability. In contrast, the anomaly boxes of CKRX and GT-HAD are lower, demonstrating a limited ability to highlight anomalies. DCAE and TDWCNet have anomaly boxes of comparable height, but TDWCNet shows a lower background box. These findings indicate that TDWCNet effectively separates anomalies from background while maintaining satisfactory background suppression and anomaly highlighting capabilities.

5) Pavia: The abnormal targets in this dataset are small in size, with a simple background, and only appear in specific localized regions. In this specific scenario, as depicted in Fig. 7(b), all methods visually appear to be capable of locating the abnormal targets in this dataset. However, traditional HAD methods, including RX, KRX, CRD, and GTVLRR, although capable of highlighting abnormal targets, exhibit poor background suppression effects, retaining a significant amount of background information such as bridges and beaches, thus increasing the false detection rate.

In contrast, CKRX and the HAD methods based on deep learning exhibit an advantage in background suppression. Although CKRX, Auto-AD, DCAE, and UADNet successfully reduce false alarms, they exhibit limited capability in highlighting anomalies, thus hindering the effective separation of abnormal targets. Additionally, they face challenges in locating anomalies, as a few abnormal targets remain unidentified. PDBSNet and GT-HAD can significantly highlight anomalies while suppressing backgrounds, but they are unreliable in locating the boundaries of abnormal targets. Conversely, although TDWCNet exhibits some false alarms in the background, it accurately locates target boundaries while highlighting anomalies, resulting in detection results closer to ground truth.

The ROC curves and PR curves of all HAD methods on this dataset are illustrated in Figs. 8(b) and 9(b), respectively. It can be observed that although the curve of GT-HAD is slightly higher than TDWCNet when the P_f value is between 0.001 and 0.01 or the recall value is between 0.7 and 0.85, TDWCNet demonstrates higher P_d and precision values in the overall curve comparison. Furthermore, as shown in Table II, TDWCNet obtains the highest $\mathrm{AUC}_{(P_d,P_f)}$ value, reflecting its capability in achieving a good balance between the detection rate and false positive rate. On the other hand, as depicted in the Box–Whisker plot in Fig. 10(b), the boxes of all methods except TDWCNet exhibit varying degrees of intersection. Although the anomaly box of TDWCNet is not the highest position, its background box is narrow enough. These findings demonstrate TDWCNet's ability to effectively suppress the background while separating anomalies.

6) San Diego I: The abnormal targets in this dataset are the three aircrafts, which are relatively large in size and concentrated in distribution. The presence of such large-size and concentrated anomalies poses higher demands on anomaly detection methods. Fig. 7(c) shows the color anomaly maps of all HAD methods on this dataset. It can be obviously observed that the detection result of TDWCNet is the most reliable because it almost completely suppresses the background and significantly highlights the abnormal targets.

Although the traditional HAD methods can detect the location of abnormal aircraft, they fail to sufficiently suppress the background, leading to a high false alarm rate, especially for KRX and GTVLRR. In comparison, PDBSNet and DCAE exhibit better background suppression effects, but they still mistakenly identify some background areas as abnormal, such as the roof of the hangar in the lower right corner. GT-HAD, while having a strong ability to highlight abnormal targets, still misdetects many background areas. On the contrary, UADNet has an excellent background suppression effect,

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but the abnormal targets it detects are incomplete. However, Auto-AD struggles to identify abnormal targets.

The corresponding ROC curves and PR curves are illustrated in Figs. 8(c) and 9(c), respectively. It is evident that although TDWCNet's curves are not always higher than those of other methods within certain ranges of P_f values and recall values, they are generally closer to the top. This demonstrates the ability of TDWCNet to maintain a high detection rate while minimizing false alarms. The fact that TDWCNet obtains the highest $\mathrm{AUC}_{(P_d,P_f)}$ value in Table II also confirms this. Additionally, it can be seen from the Box–Whisker plot shown in Fig 10(c) that there are varying degrees of overlap between the background boxes and anomaly boxes of all methods, but CKRX, PDBSNet, DCAE, UADNet, and TDWCNet have the smallest degree of overlap. Among these methods, TDWCNet exhibits the narrowest background box and the highest anomaly box, indicating its superior ability to suppress background and highlight anomalies, which is consistent with visual perception.

7) San Diego II: The ground cover categories in this dataset are the same as those in San Diego I, and the abnormal targets are also three aircrafts, but their sizes are relatively smaller. From the color anomaly maps shown in Fig. 7(d), it can be observed that the detection results of RX and Auto-AD are unsatisfactory, as although they reliably suppress the background, they barely detect any anomalies. While CRD and GTVLRR can detect abnormal targets, they fail to effectively separate anomalies from the background. KRX and GT-HAD prominently highlight abnormal targets, but GT-HAD's detected boundaries are excessively blurred, and both methods retain a considerable amount of background, resulting in a high false detection rate. Although CKRX, PDBSNet, and DCAE effectively suppress the background, their ability to highlight abnormal targets is insufficient. Compared to TDWCNet, although UADNet demonstrates superior anomaly response ability, it still retains some background structure information, resulting in false alarms. In contrast, TDWCNet not only significantly highlights anomalies while reliably suppressing the background but also accurately locates abnormal targets, making the detection result closer to ground truth.

As shown in Figs. 8(d) and 9(d), the ROC curve and PR curve of TDWCNet are almost consistently higher than those of other methods. Additionally, the quantitative analysis of the $\mathrm{AUC}_{(P_d,P_f)}$ values in Table II indicates that TDWCNet achieves the highest result among all methods. These results demonstrate that TDWCNet exhibits superior detection performance. On the other hand, as shown in the Box–Whisker plot in Fig. 10(d), except for GT-HAD, UADNet, and TDWCNet, there is overlap between the background boxes and anomaly boxes of other methods, suggesting that they struggle to separate anomalies from the background. In contrast, while TDWCNet's anomaly box may not be positioned highest, its background box is sufficiently narrow, and the gap between the anomaly and background boxes is the largest. This indicates that TDWCNet exhibits the strongest capability for separating anomalies from the background, effectively highlighting anomalies while suppressing the background.

8) Overall Comparisons: Overall, TDWCNet has exhibited exceptional performance across a diverse range of anomaly detection datasets, encompassing both point anomaly and structured anomaly scenarios. It can be visually observed from the color anomaly maps shown in Fig. 7 that, for all datasets, the background suppression effect of TDWCNet is significantly superior to other methods, almost eliminating all backgrounds. Meanwhile, TDWCNet prominently highlights the abnormal targets and its localization of anomalies is closer to the ground truth.

In the ROC curves shown in Fig. 8, although the curve of TDWCNet is not consistently above other methods at certain ranges of the p_f value, it generally appears closer to the top-left corner. This indicates that TDWCNet is capable of minimizing false positives while maintaining a high true positive rate. Similarly, in the PR curves depicted in Fig 9, while the curve of TDWCNet does not consistently outperform other methods within specific recall ranges, it typically resides closer to the upper-right corner. This illustrates TDWCNet's ability to minimize false positives from the background while accurately identifying anomalies, underscoring its effectiveness and superiority in addressing the imbalance between positive and negative samples. On the other hand, as can be observed from the Box–Whisker plots shown in Fig. 10, the distance between the background and anomaly boxes is highest for TDWCNet across all four datasets, demonstrating its satisfactory separability between background and anomalies. Furthermore, the background box of TDWCNet consistently remains the lowest and narrowest, which shows that its background suppression ability surpasses other methods. Additionally, in Table II, TDWCNet achieves the highest $\mathrm{AUC}_{(P_d,P_f)}$ values on each dataset, confirming its superior detection performance compared to other methods.

In summary, the TDWCNet is capable of adapting to various scenarios and demonstrates outstanding performance among all methods, further validating its effectiveness in the HAD task. Its capability to detect and differentiate abnormal targets of different scenes and sizes makes it a robust and versatile tool in the field of HAD.

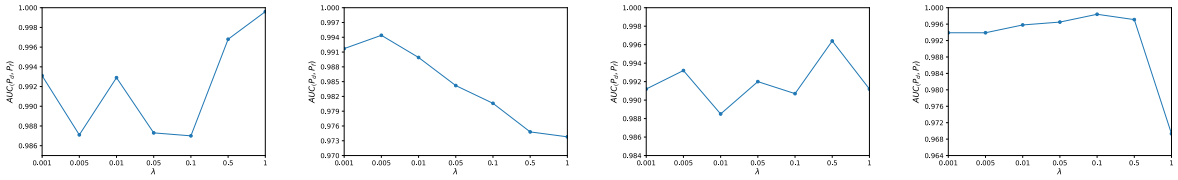
D. Parameter Analysis

In this section, we investigate the influence of hyperparameter settings, including the size of outer window w_{out} and inner window w_{in} in dual-window convolution, and the balance parameter λ in the loss function.

1) Parameters of Dual-Window Convolution: The dual-window convolution plays a crucial role in enhancing the detection performance of the model. By adjusting the windows size, the range of the receptive field can be changed to optimize the detection results. Tables III–VI display the $\mathrm{AUC}_{(P_d,P_f)}$ values of TDWCNet for detection results on the Gulfport, Pavia, San Diego I, and San Diego II datasets, with varying the size of w_{out} and w_{in}.

The results from the tables indicate that, for different datasets, despite the varying spatial sizes of the anomaly targets they contain, the optimal values of w_{out} and w_{in} tend to be relatively small. In the case of anomalies with smaller spatial extents, such as the Pavia dataset, a smaller inner

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(a) (b) (c) (d)

Fig. 11. $\mathrm{AUC}_{(P_d,P_f)}$ values for different λ taken on four datasets. (a) Gulfport. (b) Pavia. (c) San Diego I. (d) San Diego II.

TABLE III

$\mathrm{AUC}_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT w_{in} AND w_{out} VALUES TAKEN ON THE GULFPORT DATASET

w_{in} \ w_{out}	5	7	9	11	13
3	**0.9996**	0.9951	0.9741	0.9929	0.9846
5	–	0.9869	0.9763	0.9880	0.9900
7	–	–	0.9930	0.9826	0.9792
9	–	–	–	0.9869	0.9867
11	–	–	–	–	0.9847

The best results are highlighted.

TABLE IV

$\mathrm{AUC}_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT w_{in} AND w_{out} VALUES TAKEN ON THE PAVIA DATASET

w_{in} \ w_{out}	5	7	9	11	13
3	0.9852	**0.9944**	0.9883	0.9901	0.9923
5	–	0.9891	0.9932	0.9875	0.9906
7	–	–	0.9904	0.9889	0.9901
9	–	–	–	0.9935	0.9862
11	–	–	–	–	0.9882

The best results are highlighted.

TABLE V

$\mathrm{AUC}_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT w_{in} AND w_{out} VALUES TAKEN ON THE SAN DIEGO I DATASET

w_{in} \ w_{out}	5	7	9	11	13
3	0.9921	0.9883	0.9839	0.9892	0.9819
5	–	**0.9964**	0.9846	0.9888	0.9877
7	–	–	0.9915	0.9852	0.9836
9	–	–	–	0.9880	0.9831
11	–	–	–	–	0.9872

The best results are highlighted.

TABLE VI

$\mathrm{AUC}_{(P_d,P_f)}$ ↑ VALUES FOR DIFFERENT w_{in} AND w_{out} VALUES TAKEN ON THE SAN DIEGO II DATASET

w_{in} \ w_{out}	5	7	9	11	13
3	0.9395	**0.9984**	0.9867	0.9918	0.9934
5	–	0.9941	0.9955	0.9927	0.9934
7	–	–	0.9955	0.9948	0.9905
9	–	–	–	0.9939	0.9937
11	–	–	–	–	0.9919

The best results are highlighted.

window can effectively shield neighboring anomalous pixels and reconstruct a reliable background. When observing larger anomalies in the San Diego I and San Diego II datasets, although a smaller inner window may not be sufficient to completely shield related anomalies, leading to the potential reconstruction of some anomalous pixels, the Triple-UNet structure of TDWCNet can eliminate inadvertently reconstructed anomalous pixels, allowing smaller convolutional kernels to be applied to large-sized anomalies. Although a larger inner window may more effectively shield neighboring anomalous pixels, it may also result in the loss of critical background information, which is not conducive to the reconstruction of the background pixel. Furthermore, excessively large receptive fields weaken the spatial correlation between sampled information from different sub-kernels, which hinders background reconstruction.

TABLE VII

$\mathrm{AUC}_{(P_d,P_f)}$ VALUES OF ABLATION EXPERIMENTS ON FOUR DATASETS

Model	Gulfport	Pavia	San Diego I	San Diego II
Base	0.9319	0.9854	0.9629	0.9754
Base+DWC	0.9933	0.9895	0.9900	0.9834
Base+DWC+TUNet	**0.9996**	**0.9944**	**0.9964**	**0.9984**

The best results are highlighted.

2) Balance Parameter λ in Loss Function: The loss function consists of adaptive mse loss and adaptive SSIM loss. Parameter λ is used to adjust the weight of the adaptive SSIM loss, thereby controlling the influence of pixel differences and structural differences on the detection results. In the experiment, the parameter λ is selected from $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1\}$, and the settings of other parameters are consistent with those mentioned previously. Fig. 11 demonstrates the impact of different λ values on the detection results for the four datasets. As shown in Fig. 11(a), when λ is larger, the detection results exhibit higher $\mathrm{AUC}_{(P_d,P_f)}$ values, reaching the highest at $\lambda = 1$. This indicates that anomalies in Gulfport exhibit prominent structural characteristics, and the detection results primarily depend on the structural differences between the background image and the original image. For Pavia, as observed from the results depicted in Fig. 11(b), the $\mathrm{AUC}_{(P_d,P_f)}$ value reaches the highest when $\lambda = 0.005$, and gradually decreases as λ increases. This is because Pavia has a larger scale with a relatively smaller proportion of anomaly targets, making structural differences less significant as the primary difference between the background image and the original image. Therefore, selecting a smaller value of λ on Pavia to achieve optimal performance is essential. As shown in Fig. 11(c) and (d), for the San Diego I and San Diego II datasets, which contain large-sized anomalies, the $\mathrm{AUC}_{(P_d,P_f)}$ values reach the highest when λ is 0.5 and 0.1, respectively, showing that the structural loss SSIM

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TABLE VIII
TIME CONSUMPTION(S) OF DIFFERENT HAD METHODS ON FOUR DATASETS

Datasets	RX	KRX	CKRX	CRD	GTVLRR	Auto-AD	PDBSNet	DCAE	GT-HAD	UADNet	TDWCNet
Gulfport	**0.053**	169.620	11.998	9.329	72.339	27.031	180.547	27.595	22.508	23.944	175.326
Pavia	**0.043**	1059.405	151.338	4.617	119.769	32.953	246.057	70.554	47.266	25.361	265.467
San Diego I	**0.051**	173.789	16.850	8.131	70.580	26.713	175.450	29.203	21.996	28.034	187.449
San Diego II	**0.055**	177.812	8.286	7.896	72.288	25.531	174.433	21.307	22.104	24.308	205.419

plays a role in the detection process of structural anomalies. However, when λ is set to 1, an excessively high weight on the structural loss may lead to the misdetection of background objects that resemble anomalies as actual anomalies, thereby affecting the detection accuracy.

E. Ablation Study

In order to verify the effectiveness of the dual-window convolution and Triple-UNet structure in the proposed TDWCNet, multiple ablation experiments were conducted on four datasets, and the results are recorded in Table VII. Base represents using a single DWCBlock with its dual-window convolutions replaced by traditional convolutions, Base + DWC indicates the use of dual-window convolutions (at this point, the model is DWCBlock), and Base + DWC + TUNet represents the proposed TDWCNet.

It can be seen from the results in Table VII that the accuracy of anomaly detection is significantly improved after the introduction of dual-window convolution and Triple-UNet structure. The detection performance of the base model is unsatisfactory, mainly because the conventional convolution's receptive field includes the pixels to be reconstructed themselves, resulting in overly strong reconstruction capabilities, which in turn leads to the reconstruction of anomalous pixels. With the introduction of dual-window convolution, the receptive field effectively shields the pixels to be reconstructed and their neighboring correlated pixels, utilizing only background pixels far from the center for background reconstruction. Therefore, Base + DWC can obtain a purer background, enabling the detection results to achieve higher $\mathrm{AUC}_{(P_d,P_f)}$ values. However, for larger anomalies, when the inner window in dual-window convolution is smaller, the receptive field will still contain more relevant abnormal pixels, causing some anomalous pixels to be inadvertently reconstructed. In this case, the proposed TDWCNet employs the Triple-UNet structure, which organically combines three DWCBlock modules, eliminating unintentionally reconstructed anomalous pixels and obtaining a reliable background image, thereby achieving the highest $\mathrm{AUC}_{(P_d,P_f)}$ values in detection results.

F. Model Efficiency Analysis

In this section, we compare the running time (including training time and inference time) of all HAD methods on four hyperspectral datasets to illustrate the computational complexity and thus evaluate the efficiency of the models. The detailed experimental results are shown in Table VIII, recorded in seconds. The experimental results indicate that RX algorithm consistently achieves the shortest running time across all datasets, owing to its simple model design, which demonstrates its significant execution efficiency. The representation-based CRD algorithm achieves the second shortest running time. KRX method improves anomaly detection capability by mapping data into a high-dimensional feature space through a kernel method, which substantially increases computational complexity and thus the running time. CKRX is based on KRX method and groups the background pixels by clustering, which significantly reduces computational complexity. GTVLRR is time-consuming due to the iterative optimization required over numerous iterations. In contrast, deep learning-based methods are time-intensive during training because they require iterative model training. Among these methods, Auto-AD, DCAE, GT-HAD, and UADNet exhibit similar running times across the four datasets. PDBSNet, being a self-supervised method, necessitates additional operations to process raw data, thereby taking more time.

Due to the deeper network layers and the higher number of parameters in the dual-window convolution compared to conventional convolution, TDWCNet has increased computational complexity, which results in longer running times. The parameter settings of the dual-window convolution directly impact the model's computational complexity and efficiency. The extended running time is a limitation of our method, and developing lightweight HAD algorithms will be a focus of our future work. Although the efficiency of the proposed TDWCNet is not optimal, it is important to highlight that TDWCNet achieves superior detection results and demonstrates strong competitiveness.

V. CONCLUSION

In this article, we propose a novel HAD method named TDWCNet, which performs end-to-end detection in an unsupervised manner. The dual-window convolution allows the extraction of background features by leveraging the pixels between the inner and outer windows, effectively shielding the pixels to be reconstructed and their correlated pixels in the neighborhood. Based on the dual-window convolution, the DWCBlock module is constructed to extract background features for reconstructing the background image. Additionally, to address the presence of multiscale anomaly targets, the Triple-UNet structure is proposed, which organically combines three DWCBlock modules to eliminate abnormal pixels that are inadvertently reconstructed due to inappropriate convolution kernels. Furthermore, the utilization of adaptive mse loss and SSIM loss effectively suppresses the reconstruction of anomalies. Extensive experimental results demonstrate that TDWCNet exhibits excellent versatility and applicability

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across various datasets, outperforming many existing HAD methods in detection performance.

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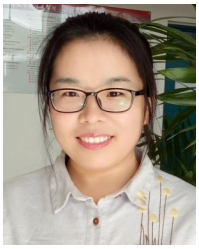
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